ELIZALDE MATHEMATICS COMPETITION

Instructions

Please write your name in CAPITALS and complete the Candidate number in the spaces below.

Use a PENCIL (B or HB)
Rub off any answers you wish to change with an eraser.

No calculators, mobile phones, smartwatches, or any electronic devices are permitted.

Show all workings clearly in the answer booklet provided.

Maintain **silence** during the exam. Raise your hand if you need assistance.

Candidate Name	
Country	NIGERIA
Class	YEAR 9
CRC	EMC25001
Candidate Roll No	

NATIONAL ROUND

1) Ella wants a rectangle with
   - A perimeter of 100cm and
   - The largest possible area

   What are the dimensions of the rectangle that satisfies her condition?

Identify the following equation as being linear or non-linear. (Question 3-4)

2) $3x + 2y = 7$
3) $x^2 - 3x + 4 = y$
4) $9x - 2 = \frac{y}{2}$

5) Find the length of the diagonal of a rectangle, D given that the length of the rectangle is 10.7cm and the diagonal makes an angel of 39^0 with the longer side. Correct your answer to 1 decimal place
6) Adams earns £5, 000 per week. Each week, he spends £3x on food, £2x on rent and saves the rest. How much does he save in a week?

Demi charges £12 to cut and rake one lawn. She earned £1176 for the summer. (Question 7-8)

7) Model this situation using an equation.
8) Solve the equation to determine how many lawns she cut and raked over the summer

9) $\angle 1$, $\angle 2$, and $\angle 3$ are adjacent angles, with $\angle 1$ supplementary to $\angle 2$ and $\angle 2$ complementary to $\angle 3$. If $M\angle 1 = (8x + 3)°$ and $M\angle 2 = (5x - 18)°$, find $M\angle 3$

Let $A = \{1, 2, 3, 8, 10, 12\}$ and $B = \{6, 5, 4, 3, 2, 10\}$ write down the set.

10) $A \cap B$

11) $A \cup B$

12) Solve the inequality $3x - 7 > 11$. Write your answer using set notation

13) If $(A + B)^2 = A^2 + 2AB + B^2$ then what can we say about A and B? (Assume AB and BA exists)

14) If the sides of a right triangle are in the ratio 3:4:5, and the shortest side is 9cm, find the length of the hypotenuse.

15) If A is a n x m matrix such that AB and BA are both defined, then order of B is?

16) A shopkeeper purchased 200 bulbs for £10 each. However 5 bulbs were fused and had to be thrown away. The remaining were sold at £12 each. Find the gain or loss%.

17) Find the sine of an angle *a* in a right triangle, where the opposite sides is 6 units and the hypotenuse is 10 units.

18) The price of a motor bike was £60, 000 last year. It has increased by 15% this year. What is the price now?

$\xi = \{103, 104, 105, 109, 110, 112, 114\}$

A = even number from ξ

$B = \{103, 112, 114\}$

State which numbers belong to the following:

19) $A \cup B$

20) A'